

Muhammad Ali

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EDUCATION

William & Mary

Williamsburg, VA

Bachelor of Science in Computational & Applied Mathematics & Statistics

EXPERIENCE

William & Mary

Williamsburg, VA

Teaching Assistant – MATH 451/551: Probability & MATH 452/552: Mathematical Statistics

SELECTED COURSEWORK

Mathematics

MATH 451/551 – Probability
MATH 452/552 – Mathematical Statistics
MATH 455/555 – Statistical Learning
MATH 311 – Real Analysis
MATH 403 – Measure Theory (*planned*)
MATH 405 – Complex Analysis

Computer Science

CSCI 416/516 – Fundamentals of AI/ML
CSCI 446/546 – Neural Networks for ML
CSCI 436/536 – Data Mining (*planned*)
CSCI 456/556 – Large Language Models (*planned*)
CSCI 420/520 – High-Dimensional Geometry of Data
CSCI 455/555 – GenAI for Software Engineering (*planned*)

PROJECTS

Individual Brown Bear Classification | *Python, TensorFlow/Keras, ResNet-50, Computer Vision*

- Developed an individual-bear image classifier as part of a larger computer-vision pipeline for wildlife monitoring using camera-trap data from Kenai National Wildlife Refuge.
- Constructed a model-ready dataset and used five-fold stratified group cross-validation by source video to reduce data leakage between highly correlated training and validation frames.
- Evaluated a transfer-learning pipeline using an ImageNet-pretrained ResNet-50 with data augmentation, hyperparameter experimentation, model evaluation, and dataset/model documentation.

Persistent Homology of Sampled Manifolds | *Python, Jupyter, Topological Data Analysis*

- Conducted a computational topology experiment studying how persistent homology recovers the topology of a sinusoidal manifold under varying sample sizes, noise levels, and sampling distributions.
- Generated point clouds, constructed Vietoris–Rips filtrations, and analyzed persistence features to distinguish underlying topological structure from artifacts of finite sampling and noise.
- Organized reproducible experiments while tracking computational performance, including runtime, memory usage, experimental parameters, and generated persistence results.

Machine Learning Portfolio Website | *HTML, CSS, JavaScript, GitHub Pages*

In Progress

- Designing and maintaining an interactive technical portfolio organizing work across mathematics, machine learning, algorithms, and software engineering.
- Developing visual explanations of mathematical concepts underlying machine learning and publishing technical projects through GitHub Pages.

TECHNICAL SKILLS

Languages: Python, R, C++, Java, LaTeX

Tools: Git, GitHub, Jupyter Notebook, VS Code

ML / Data Libraries: NumPy, pandas, Matplotlib, scikit-learn, TensorFlow/Keras